Transfusion on-call: A haematologist's cheat sheet

Transfusing patients with red cell antibodies

Australian Red Cross Lifeblood®

Table 1: Red cell selection based on the urgency of transfusion

Category	Action
Emergency	Transfuse emergency issue group O uncrossmatched RBC as per hospital or laboratory policy. Never transfuse based on a historical ABO blood group. Antigen negative blood should be given, where possible, to patients with historical red cell antibodies.
Urgent (<1 hr)	Transfuse group-specific, IAT crossmatch compatible, antigen-negative RBC (where available).
Routine	Transfuse group-specific, IAT crossmatch compatible, extended phenotype matched RBC.

Table 2: Red cell selection for routine transfusion when the red cell antibody hasn't been identified but transfusion must proceed

Scenario	Differential diagnosis	Transfusion strategy
Variable reaction strengths	Multiple red cell antibodies	ABO RhD compatible extended phenotype matched RBC
Positive auto in the 11 cell panel **and** biochemical evidence of haemolysis	Red cell autoantibody	ABO RhD compatible extended phenotype matched RBC
Positive auto in the 11 cell panel, biochemical evidence of haemolysis **and** recent transfusion	Red cell alloantibody (DHTR)	ABO RhD compatible extended phenotype matched RBC
Weak reaction strengths (<2+)	High titre low avidity antibody Antibody to a high frequency antigen	ABO RhD compatible phenotype matched RBC with the weakest IAT crossmatch reactions, commonly referred to as "least incompatible". Note: Occasionally crossmatch compatible units can be found prior to determining the specificity of the red cell alloantibody.

Transfusing patients with red cell antibodies

EMERGENCY (LIFE-SAVING) TRANSFUSION

- Transfuse group O uncrossmatched emergency issue red cells as per local hospital or laboratory policy.
- NEVER transfuse a patient based on a historical ABO blood group.

National statement for the emergency use of group O uncrossmatched RBC

- Emergency uncrossmatched group O RBC are to be used only in an emergency to save a patient's life and when there is no current valid pretransfusion specimen, not as a time-saving measure.
- Where a patient's ABO RhD blood group is unknown or has not been confirmed by current pretransfusion testing then the following applies:

  For females of childbearing potential (≤50 years) and paediatric males ≤18 years (or as per local paediatric policy)
  Issue **group O RhD negative** uncrossmatched RBC until the patient's ABO RhD blood group has been determined using a valid current specimen.

  For females > 50 years and all adult males >18 years
  Issue **group O RhD positive** uncrossmatched RBC until the patient's ABO RhD blood group has been determined using a valid current specimen.
- The patient's ABO group MUST be determined as soon as possible to allow ABO RhD group specific RBCs to be issued.
- When four units of uncrossmatched group O RhD negative RBC have been issued to a patient of unknown RhD group, transfusion of group O RhD positive RBC should be considered irrespective of age, gender or child-bearing potential.

- Antigen negative blood should be considered in patients with known historical clinically significant red cell alloantibodies
- A historical red cell phenotype can be used to select RBC for emergency transfusion
- The risk of haemolysis with transfusion of uncrossmatched RBC is ~ 0.1%
- The haemolysis is generally limited to biochemical changes and NOT associated with a clinically significant acute transfusion reaction

Why?

- Most red cell alloantibodies are IgG, which cause extravascular haemolysis
- The implicated alloantibody in plasma is also lost in massive haemorrhage

Reference: Yazer et al. Use of Uncrossmatched Erythrocytes

Clinically significant red cell alloantibodies

Red cell alloantibodies capable of causing haemolysis at 37°C.

Significance and action	Red cell alloantibodies
Always clinically significant Red cells for transfusion should be Antigen negative and IAT crossmatch compatible at 37°C	Anti A, B, H Anti D, C,c, E,e Anti K, k Anti Fya, Fyb Anti Jka, Jkb Anti S, s, U
Only significant if active at 37°C Red cells for transfusion should be IAT XM compatible at 37°C	Anti M (should be antigen negative) Anti N Anti Kpa Anti Lea, Leb, Anti Coa, Cob
Rarely significant Red cells for transfusion should be least reactive by IAT XM	HTLA

Source: ANZSBT Guidelines for transfusion and immunohaematology laboratory practice

Transfusing patients with red cell antibodies

URGENT TRANSFUSION (REQUIRED IN <1 HR)

Depending on the time available and the ease of identifying the red cell antibody, RBC for urgent transfusion should be (in order of decreasing priority):

- Group specific, IAT crossmatch compatible, antigen negative
- Group specific, IAT crossmatch compatible
- Group specific, uncrossmatched

Note: ABO compatible RBC can be transfused when group specific RBC are not available in hospital inventory.

Important: Where transfusion is urgently required, particularly in life-threatening situations, it may be necessary to provide ABO/RhD compatible but otherwise serologically incompatible red cells until further investigations are completed. The treating clinical team must be informed.

Source: ANZSBT Guidelines for transfusion and immunohaematology laboratory practice

ROUTINE TRANSFUSION

When the red cell alloantibody has been identified

RBC for transfusion should be group specific, IAT crossmatch compatible, antigen negative, and extended phenotype matched where appropriate.

An extended phenotype generally includes typing of the Rh, Kell, Duffy, Kidd, S,s blood group systems.

If extended phenotype matched RBC for transfusion are not available. It is suggested to omit antigens in the following order: Ss, Jk, Fy, K, Rh.

Red cell phenotyping can NOT be performed in patients who have been transfused in the past three months as transfused RBC will be present in the sample. The sample should be sent to Australian Red Cross Lifeblood for genotyping.

When the red cell antibody has NOT been identified but transfusion is required

Important: Confirm with the treating clinician if transfusion can be delayed. If transfusion can not be delayed refer to Table 2.

Multiple red cell alloantibodies with specificity still to be determined

Clues

- Variable reaction strength
- Often occasional negative cells
- Negative auto control

Approach

- Transfuse ABO RhD compatible extended phenotype matched RBC
- If the local laboratory cannot determine the antibody or antibodies specificity the sample should be sent to the local reference laboratory or Lifeblood Red Cell Reference Laboratory for further investigation.

Transfusing patients with red cell antibodies

Panagglutination

Clues

- All cells positive in three cell screen and antibody panels
- Auto control may be negative or positive
- Reactions may be weak (<2+), variable (1-2+) or strong (3-4+) and generally similar

Causes

- High titre low avidity antibody (antibody present in high concentrations but binds weakly to red cell antigens)
- Antibody to a high frequency antigen (antigen generally present in >90% individuals)
- Reagent related antibody, i.e. antibodies against preservatives in reagent RBC
- Rouleaux
- Warm or cold autoimmune haemolytic anaemia

Approach

- In these scenarios antibody identification in hospital laboratories may be challenging.
- If the specificity of the alloantibodies cannot be determined, the sample should be sent to the local reference laboratory or Lifeblood Red Cell Reference Laboratory for further investigation.
- **If transfusion can NOT be delayed, transfuse ABO RhD compatible phenotype matched RBC with the weakest IAT crossmatch reactions, commonly referred to as "least incompatible".**
- Occasionally crossmatch compatible units can be found prior to determining the specificity of the red cell antibody.

Warm or cold autoimmune haemolytic anaemia

- If panagglutination occurs in the context of a positive auto control AND biochemical evidence of haemolysis, autoimmune haemolytic anaemia should be suspected. Refer to the *The Direct Antiglobulin Test* cheat sheet.

Note: If a patient has been transfused in the past three months, a positive auto control (and DAT) can be due to an alloantibody against transfused RBCs as the sample contains both patient's own RBC AND residual transfused RBC.

Antibody screen flowchart

